Number-Theoretic Functional Equations

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1 Examples

IMO Shortlist 2015 A2

Example 1: Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ with the property that

$$f(x - f(y)) = f(f(x)) - f(y) - 1$$

holds for all $x, y \in \mathbb{Z}$.

Proof: $y = f(x) \implies f(x - f(f(x))) = -1$

$y = x - f(f(x)) \implies f(x + 1) = f(f(x))$

$x = f(y) - 1 \implies f(-1) + 1 = f(f(f(x) - 1)) - f(x) = f(f(x)) - f(x) = f(x + 1) - f(x)$

So, $f = mx + c$, and matching coefficients,

$m^2 = m, 2mc = c + 1$

So, $(m, c) = (0, -1), (1, 0)$

$f(x) \equiv -1$ or $f(x) = x + 1$

AoPS solution: me

ELMO 2025 P1

Example 2: Let $\mathbb{Z}_{>0}$ denote the set of positive integers. Find all functions $f \colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that for all positive integers m and n,

$$f^m(n) + f(mn) = f(m)f(n).$$

Note: $f^m(n) = \underbrace{f(f(\cdots f}_{m \text{ times}}(n)\cdots))$, that is, f applied m times to n.

Proof: $P(1, n) \implies f(1) = 2$

$P(m, 1) \implies f^m(1) = f(m) \ldots (1)$

Taking f on both sides gives $f^{m+1}(1) = f(f(m))$

Now, we take 2 cases.

Case 1: f is injective.

$f^{m+1}(1) = f(m + 1)$ by (1), so $f(f(m)) = f(m + 1) \implies f \equiv m + 1$.

Case 2: f isn't injective.

Say $f(x) = f(y)$ for some $x \neq y$.

$f^x(1) = f^y(1)$, so f is periodic from x. Thus, f is bounded. Let the upper bound be C.

By periodicity, we can choose a, b such that $f(a) = f(b) = C$. Then, $P(a,b) \implies 2C \leq f^a(b) + f(ab) = C^2$.

Thus, $C = 2$. If $f(k) = 1$ for some k, then $P(k,k)$ gives a clear contradiction.

Thus, our solutions are $\boxed{f \equiv 2}$ and $\boxed{f \equiv m+1}$.

AoPS solution: me

INMO 2024 P4

Example 3: A finite set $\mathcal{S}$ of positive integers is called cardinal if $\mathcal{S}$ contains the integer $|\mathcal{S}|$ where $|\mathcal{S}|$ denotes the number of distinct elements in $\mathcal{S}$. Let f be a function from the set of positive integers to itself such that for any cardinal set $\mathcal{S}$, the set $f(\mathcal{S})$ is also cardinal. Here $f(\mathcal{S})$ denotes the set of all integers that can be expressed as $f(a)$ where $a \in \mathcal{S}$. Find all possible values of $f(2024)$.

Proof: $S = \{1\} \implies f(1) = 1$

$S = \{1,2\} \implies f(2) \in \{1,2\}$

If $f(2) = 1$, $S = \{2, 2024\} \implies f(2024) \in \{1,2\}$.

From now, assume $f(2) = 2$, $f(2024) \neq 1, 2$

If $f(3) = 2$, $S = \{2,3\} \implies \{2,2\}$ is cardinal, $\Rightarrow\Leftarrow$.

If $f(3) = 1$, $S = \{1,3,2024\} \implies \{1, f(2024)\}$ is cardinal, so $f(2024) = 1, 2$, $\Rightarrow\Leftarrow$.

$S = 1,2,3 \implies 1, 2, f(3)$ is cardinal, so $f(3) = 3$.

Now, we claim $f(n) = n \; \forall \; n \in \mathbb{N}$.

Say $f(1) = 1, f(2) = 2, \ldots, f(k) = k$.

FTSOC $f(k+1) \leq k$.

$S = \{1, 2, \ldots, a-2, a, k+1\} \implies \{1, 2, \ldots, a-2, a, f(k+1)\}$ is cardinal, so $f(k+1) \neq 1, 2, \ldots, a-2, a \; \forall \; 2 \leq a \leq k$.

And, $S = \{1, 2, \ldots, k, k+1\} \implies \{1, 2, \ldots, k, f(k+1)\}$ is cardinal, so $f(k+1) = k+1$.

Thus, $\boxed{f(2024) \in \{1, 2, 2024\}}$.

AoPS solution: me

India IMOTC 2025, Day 4, P2

Example 4: Does there exist a function $f : \{1, 2, \ldots, 2025\}^2 \to \{1, 2, \ldots, 2025\}$ such that:
- for any positive integer $i \leqslant 2025$, the numbers $f(i,1), f(i,2), \ldots, f(i, 2025)$ are all distinct, and
- for any positive integers $i \leqslant 2025$ and $j \leqslant 2024$, $f(f(i,j), f(i,j+1)) = i$?

Proof: Since $f(i,1), f(i,2), \ldots, f(i,2025)$ are distinct, $\exists \; k$ such that $f(f(i,i),k) = i$

Notice that $f(f(f(i,i),k), f(f(i,i),k+1)) = f(i, f(f(i,i),k+1)) = f(i,i)$

So, $f(f(i,i),k+1) = i$, $k = 2025$.

Say $f(i,i) = i$. Then, $f(i, f(i,i+1)) = i$, so $i = 2025$ or $\Rightarrow\Leftarrow$.

Say $f(i,i) = f(j,j)$. Then, $i = f(f(i,i),2025) = f(f(j,j),2025) = j$.

So, $\exists\ i$ such that $f(i,i) = 2024$. Then, $f(2024,2025) = 2024$ but $f(f(2024,2024), f(2024,2025)) = 2024 \implies f(2024,2025) = 2025$, $\Rightarrow\Leftarrow$.

AoPS solution: me

INMO 2026 P4

Example 5: Two integers a and b are called companions if every prime number p either divides both or none of a, b. Determine all functions $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that $f(0) = 0$ and the numbers $f(m) + n$ and $f(n) + m$ are companions for all $m, n \in \mathbb{N}_0$.
(Here $\mathbb{N}_0$ denotes the set of all non-negative integers.)

Proof: Solution from the contest:

Denote the assertion by $P(m,n)$.

$P(m,0) \implies \text{rad}(m) = \text{rad}(f(m))$. Denote this assertion by $Q(m)$.

Claim 1: $f(1) = 1$.

Proof: $Q(1) \implies \text{rad}(f(1)) = \text{rad}(1)$, so $f(1) = 1$. $\square$

Claim 2: $f(p^k) = p^k$.

Proof: $Q(p^k)$ gives $f(p^k) = p^\ell$, and $P(p^k, 1)$ gives $\text{rad}(p^k + 1) = \text{rad}(p^\ell + 1)$, so $k = \ell$ by Zsig. $\square$.

Claim 3: $f \equiv m$

Proof: Notice that $\gcd(p^k + m, p^k + f(m)) = \gcd(p^k + m, f(m) - m) \geq \text{rad}(p^k + m)$.

Since $\text{rad}(p^k + m)$ is unbounded, plugging in large enough p, k gives $f(m) = m$ as desired. $\square$

AoPS solution: me

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Example 6: Let $\mathbb{N}$ denote the set of positive integers. A function $f : \mathbb{N} \to \mathbb{N}$ is said to be *bonza* if

$$f(a) \text{ divides } b^a - f(b)^{f(a)}$$

for all positive integers a and b.
Determine the smallest real constant c such that $f(n) \leqslant cn$ for all bonza functions f and all positive integers n.

Proof: Notice that $P(a,a) \implies f(a) \mid a^a$. Also, $f \equiv a$ works, so from now assume f isn't the identity. So, $\exists\ y$ such that $f(y) \neq y$.

Claim 1: $f(p) = 1$ for large enough p.

Proof: $P(p,y) \implies f(p) \mid y - f(y)$, contradiction for large enough p.

Claim 2: $f(2m+1) = 1$

Proof: Let q be a prime such that $q \equiv 2 \pmod{p}$ and $f(q) = 1$, which exists by Dirichlet.

$P(p,q) \implies p \mid f(p) \mid q^p - 1$, contradiction.

So, f has no prime factors apart from 2 for an odd input, and $f(2m+1) \mid (2m+1)^{2m+1}$, so $f(2m+1) = 1$.

Claim 3: $f(x) = 2^k$

Proof: Say $r \mid f(x)$. Then, $r \mid f(x) \mid r^x - 1$, contradiction

Now, $P(x,3) \implies f(x) \mid 3^x - 1$, so $\nu_2(f(x)) \leq \nu_2(3^x - 1) = \nu_2(x) + 2$.
Combining with claim 3, we get $f(x) \leq 4x$, which can easily be constructed.

AoPS solution: me

2 Practise Problems

Here are some practise problems, in increasing order of difficulty.

1. **IMO Shortlist 2019 A1**
   Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that, for every pair of integers a, b,
   $$f(2a) + 2f(b) = f(f(a+b)).$$

2. **IMO Shortlist 2014 A4**
   Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ satisfying
   $$f(f(m) + n) + f(m) = f(n) + f(3m) + 2014$$
   for all integers m, n.

3. **IMO Shortlist 2013 A5**
   Find all functions $f : \mathbb{Z}_{\geq 0} \to \mathbb{Z}_{\geq 0}$ satisfying
   $$f(f(f(n))) = f(n+1) + 1$$
   for every non-negative integer n.

4. **IMO Shortlist 2015 A5**
   Let $2\mathbb{Z} + 1$ denote the set of odd integers. Find all functions $f : \mathbb{Z} \to 2\mathbb{Z} + 1$ satisfying
   $$f(x + f(x) + y) + f(x - f(x) - y) = f(x + y) + f(x - y)$$
   for all integers x, y.

5. **IMO Shortlist 2019 N4**
   A positive integer constant C is given. Find all functions $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that
   $$a + f(b) \mid a^2 + bf(a)$$
   for all positive integers a, b satisfying $a + b > C$.

6. **IMO Shortlist 2014 A6**
   Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that
   $$n^2 + 4f(n) = f(f(n))^2$$
   for every integer n.